

# Hammad Ahmad

AI / Machine Learning Engineer | LLMs · RAG · Knowledge Graphs

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#) | [ORCID](#)

## PROFILE

AI/ML engineer with an MSc (Merit) in Applied AI and a first-author Springer publication, focused on production LLM systems. I build retrieval-augmented pipelines pairing vector search with knowledge graphs to keep answers grounded and citable, one reaching 0.82 source accuracy and 0.81 citation quality, and take work from retrieval design through backend APIs to deployment, including a voice-agent backend whose call latency I cut 54%.

## TECHNICAL SKILLS

**Machine Learning, NLP & LLMs** : PyTorch, TensorFlow, scikit-learn, XGBoost, LightGBM, Retrieval-Augmented Generation (RAG), Sentence Transformers, vector embeddings, semantic search, cross-encoder re-ranking, RAGAS evaluation, Ollama

**Engineering & Data** : Python, TypeScript, JavaScript, C++, SQL, FastAPI, Flask, REST APIs, React, PostgreSQL, Neo4j, pandas, NumPy

**MLOps, Cloud & DevOps** : MLflow, Docker, Git, GitHub Actions (CI/CD), AWS (EC2, S3), GCP (Cloud Run), Linux

## EXPERIENCE

### Outlyst

Oct 2025 - Mar 2026

AI / Machine Learning Engineer (fixed-term contract)

Leeds, UK (Remote)

- Built and deployed an AI voice-agent system for outbound calling (Retell AI, FastAPI): backend dialogue-flow logic, automated call flows and lead-qualification rules, handling 2,100+ calls and cutting manual calling workload by roughly 50%.
- Reduced average call latency 54%, from 2.4s to 1.1s, through async callbacks, connection pooling and backend optimisation, materially improving conversational responsiveness.
- Built an internal micro-CRM for lead tracking, removing external CRM licensing costs.

### University of Bradford

Jan 2025 - Sep 2025

Research Assistant, Machine Learning & LLMs

Bradford, UK

- Built **FinLaw-UK** (code + benchmark), a RAG system pairing Mistral 7B (served locally via Ollama) with a Neo4j knowledge graph for UK financial-regulation QA, grounded in FCA, PRA, FRC and statutory sources.
- Engineered the retrieval pipeline: clause-level segmentation, Sentence Transformer embeddings, cross-encoder re-ranking, and Neo4j citation checks flagging missing references as hallucinations.
- Evaluated with RAGAS and custom metrics: 0.82 source accuracy, 0.81 citation quality, improving on a standalone LLM baseline.

### COMSATS University Islamabad

Jul 2023 - Jul 2024

Research Assistant, Data Science

Islamabad, Pakistan

- Benchmarked 11 classifiers for diabetes risk on a 253,680-record CDC BRFSS dataset (14% positive class), comparing random over-sampling, SMOTE and ADASYN with resampling confined to training folds; Random Forest performed best on ROC-AUC and sensitivity.

## PROJECTS

### Jobzyl: Multi-Platform Job Aggregator

[jobzyl.com](#)

- Live platform fanning one query out to 20 job-board APIs across 60+ countries in parallel, deduplicating server-side and streaming results over Server-Sent Events (first results ~1.4s).
- Built an ATS-style CV matcher (browser-side keyword extraction and overlap scoring), Kanban application tracker and CSV/JSON export; React and TypeScript, CV text encrypted at rest.

## EDUCATION

### University of Bradford

Sep 2024 - Sep 2025

MSc, Applied Artificial Intelligence and Data Analytics (Merit)

Bradford, UK

- **Dissertation**: FinLaw-UK: A Graph-Augmented Retrieval Chatbot for Reliable and Transparent UK Financial Regulation.

### COMSATS University Islamabad

Sep 2020 - Jul 2024

BS, Bioinformatics

Islamabad, Pakistan

## PUBLICATIONS

Ahmad, H., Khan, M.U., & Azam, M. (2024). Comparative Analysis of Machine Learning Methods for Enhancing Sleep Efficiency and Prediction. *ICSMAI 2024* (Springer). DOI: [10.1007/978-3-031-66854-8\\_1](#)

**LANGUAGES**: English (IELTS 7.0, CEFR C1) | Urdu (Native) | German (A1.2)

**CURRENTLY**: Quality Control Assistant, Myton Food Group (Morrisons), Bradford, Apr 2026 - Present, while pursuing AI/ML roles and PhD applications.